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AIRCRAFT STRUCTURAL INTEGRITY PROGRAM AND AIR AND SPACE EQUIPMENT STRUCTURAL MANAGEMENT

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This publication implements the integrity program requirements related to structural integrity in Air Force Policy Directive (AFPD) 63-1/20-1, *Integrated Life Cycle Management,* and is consistent with DoD Directive 5000.01, *The Defense Acquisition System,* and DoD Instruction 5000.02T, *Operation of the Defense Acquisition System.* It also provides structural management direction associated with AFPD 21-1, *Maintenance of Military Materiel*, and designates offices and procedures that provide Air Force technical and engineering support for air and space assets utilizing technologies in the structural disciplines. This Department of the Air Force Instruction

(AFI) specifically addresses the structural disciplines of advanced composites, corrosion prevention and control, low observables supportability, metals technology, and nondestructive inspection (commonly abbreviated as "NDI") for non-facilities assets. This publication applies to all military and civilian employees of the Regular Air Force, Air Force Reserve, and Air National Guard, and to other individuals or organizations as required by binding agreement or obligation with the Department of the Air Force. Any contractor requirements contained within this instruction must be contained within the contract, grant, or agreement to be enforceable. Ensure all records created as a result of processes prescribed in this publication are maintained in accordance with AFI 33-322, *Communication and Information Records Management and Information Governance Program,* and disposed of in accordance with the Air Force Records Disposition Schedule located in the Air Force Records Information Management System. Refer recommended changes and questions about this publication to the Office of Primary Responsibility (OPR) using the AF Form 847, *Recommendation for Change of Publication*. Route AF Forms 847 from the field through the appropriate functional chain of command. This publication may be supplemented at any level, but to ensure standardization, any organization supplementing this instruction must send the implementing publication to the Engineering & Force Management Division, office of the Deputy Assistant Secretary for Science, Technology and Engineering (SAF/AQRE) for review and coordination before publishing. The authorities to waive wing/unit level requirements in this publication are identified with a Tier ("T-0," "T-1," "T-2," "T-3") number following the compliance statement. See AFI 33-360, *Publications and Forms Management,* for a description of the authorities associated with the Tier numbers. Submit requests for waivers through the chain of command to the appropriate Tier waiver approval authority, or alternately, to the requestor's commander for non-tiered compliance items. Tiering is further addressed in **paragraph 1.4**.

(AFMC) This publication supplements and implements the guidance of Department of the Air Force Instruction (DAFI) 63-140, *Aircraft Structural Integrity Program and Air and Space Equipment Structural Management*. This supplement adds the roles and responsibilities of the Air Force Life Cycle Management Center (AFLCMC), Air Force Research Laboratory (AFRL), and Air Force Sustainment Center (AFSC) in **paragraph 2** and expands requirements relating to structural disciplines in **paragraph 4** This supplement applies to AFMC Regular Air Force (RegAF), the Air National Guard (ANG), the Air Force Reserve Command (AFRC) and their units and the United Space Force (USSF) organizations for application to Space Systems and Equipment. This publication may be supplemented, but supplements must be provided to the Office of Primary Responsibility (OPR) of this publication for review prior to publication. The authorities to waive wing/Space Force equivalent/unit level requirements in this publication are identified with a Tier ("T-0, T-1, T-2, T-3") number following the compliance statement. See Department of the Air Force Manual (DAFMAN) 90-161, *Publishing Processes and Procedures*, for a description of the authorities associated with the Tier numbers. Submit requests for waivers using DAF Form 679, *Air Force Publication Compliance Item Waiver Request/Approval*, through the chain of command to the appropriate Tier waiver approval authority, or alternately, to the Publication OPR for non-tiered compliance items. All requests for waiver must be approved by the publication OPR prior to implementation. Refer recommended changes and questions about this publication to the office of primary responsibility (OPR) using the DAF Form 847, *Recommendation for Change of Publication*; route DAF Forms 847 from the field through the appropriate functional chain of command. Ensure that all records created as a result of processes

prescribed in this publication are maintained IAW Air Force Instruction (AFI) 33-322, *Records Management and Information Governance Program*, and disposed of IAW the Air Force Records Disposition Schedule located in the Air Force Records Information Management Systems). Compliance with the attachments in this publication is mandatory.

SUMMARY OF CHANGES

This document has been substantially revised. Major changes include the integration and streamlining of content from AFI 20-114. The integration removes Air Force Materiel Command (AFMC) detailed procedures previously in AFI 20-114. Removed procedures may be addressed in a supplement or another lower-level Air Force publication.

(AFMC) This new supplement to DAFI 63-140 includes the integration of the content and procedures from AFI 20-114, as discussed above, and includes additional content from AFI 20-114 AFMC Supplement. This includes the roles and responsibilities and expands the structural disciplines requirements.

SUMMARY OF CHANGES CORRECTIVE ACTIONS

This corrective action corrects the page count for the AFMC supplement. It has been updated on 13 February 2023.

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1. Overview.

1.1. Publication Organization.

1.1.1. This instruction integrates directive guidance for the Air Force Aircraft Structural Integrity Program (commonly abbreviated as “ASIP”) with directive guidance for the complementary air and space equipment structural management disciplines.

1.1.2. **Paragraph 2** and its subparagraphs contain combined roles and responsibilities. Where appropriate, this instruction clearly delineates separate Aircraft Structural Integrity Program and structural management roles and responsibilities.

1.1.3. **Paragraph 3** and its subparagraphs contain specific directive guidance and procedures for the Aircraft Structural Integrity Program.

1.1.4. **(Added-AFMC) Paragraph 4** and its subparagraphs contain specific directive guidance and Requirements as Relating to Structural Disciplines. All support, to include engineering assistance and technical support, described in this paragraph will be also provided to USSF organizations for application to Space Systems and Equipment.

1.2. **Air Force Aircraft Structural Integrity Program.**

1.2.1. The Aircraft Structural Integrity Program requires aircraft program offices and Major Commands (MAJCOM) to collaborate on a series of time-phased tasks during the development, acquisition, production, modification, and sustainment of Air Force aircraft. These collaborative tasks support aircraft mission readiness and airworthiness assurance.

1.2.2. The goal of the Aircraft Structural Integrity Program is to ensure the structural safety, performance, durability, and supportability of Air Force aircraft throughout their service lives, with the least possible economic burden.

1.2.2.1. The objectives of the Air Force Aircraft Structural Integrity Program are to:

1.2.2.2. Define the structural integrity requirements necessary to support airworthiness assurance.

1.2.2.3. Establish, evaluate, substantiate, and certify the structural integrity of aircraft structures.

1.2.2.4. Acquire, evaluate, and apply aircraft usage and maintenance data to ensure the continued structural integrity of operational aircraft.

1.2.2.5. Provide quantitative information for decisions on force structure planning, inspection, modification priorities, and risk management. The Aircraft Structural Integrity Program also provides information on expected life cycle costs and related operational and support issues.

1.2.2.6. Provide a basis for improving structural criteria and methods of design, manufacturing, evaluation, and substantiation for future aircraft systems and modifications.

1.3. **Air and Space Equipment Structural Management.**

1.3.1. This publication addresses the structural management disciplines of advanced composites, corrosion prevention and control, low observables supportability, metals technology, and nondestructive inspection (commonly abbreviated as “NDI”). Structural management disciplines complement the Aircraft Structural Integrity Program. They also optimize safety, serviceability and readiness throughout the life cycle of air and space equipment.

1.3.1.1. **(Added-AFMC)** This publication adds the structural management discipline of metals technology for aircraft structural maintenance commonly known as sheet metal.

1.3.2. Structural management, as addressed in this publication, refers to the technical and engineering support activities associated with requirements development, design, maintenance, and integrity monitoring-assessment of air and space physical structural equipment and assets. AFMC maintains offices with structural management technical experts. These experts advise MAJCOMs, program managers, and the AFMC enterprise

on the structural management disciplines. They provide engineering and technical support on the maintenance, training, supportability, and development of metal, plastics, bonded honeycomb and advanced composite repair capabilities. They also support Air Force-wide implementation of facility, tooling, processing and equipment guidelines. The AFMC structural management technical experts may also advise or review facilities and infrastructure projects. However, this publication only addresses the structural management requirements and responsibilities for non-facilities Air Force air and space equipment. **Note:** Specific directive guidance for the structural maintenance career fields, maintenance technicians, and maintenance specialties is included in AFI 21-101, *Aircraft and Equipment Maintenance Management.* AFI 63-140 and the structural maintenance guidance in AFI 21-101 are complementary.

1.3.2.1. Advanced composites are composed of various filament reinforcements of high strength, stiffness, electrical, chemical or other specialized properties embedded in matrix materials with tailored physical properties. AFMC advanced composites technical experts provide engineering and technical support on the repair, maintenance, training, development, testing, and selection of advanced composites. They also support Air Force-wide implementation of facility, tooling, processing and equipment guidelines for advanced composites supportability processes in accordance with Technical Order (TO) 1-1-690, *General Advanced Composite Repair Processes Manual.*

1.3.2.2. Corrosion prevention and control is the materials, processes, and activities utilized to combat the deterioration of a material or its properties due to the reaction of that material with its chemical environment. AFMC corrosion prevention and control technical experts provide engineering and technical support to the Air Force Corrosion Control and Prevention Executive (a position required in each Military Service by Title 10 United States Code Section 2228, *Office of Corrosion Policy and Oversight*), in addition to program managers and MAJCOMs, for corrosion prevention, mitigation, and control on air and space systems Air Force-wide.

1.3.2.3. Low observables are the technologies implemented to control the radar, infrared, visual, and acoustic signatures in aircraft to reduce their vulnerability. AFMC low observables technical experts provide engineering and technical support to ensure the integrity, reliability and maintainability of low observable materials, repair processes, nondestructive inspections, signature diagnostic measurements, and signature health assessments for air and space systems Air Force-wide.

1.3.2.4. Metals technology encompasses the technologies used in the manufacture, fabrication, heat treatment, welding, inspection, and testing of metal parts for aircraft and aerospace equipment. AFMC metals technology technical experts provide engineering and technical support for additive manufacturing (i.e., 3-dimensional printing) and support its implementation Air Force-wide. They establish Air Force-wide guidance for welding certification, and advise the AFMC enterprise and MAJCOMs on aerospace welding repairs and processes, as requested.

1.3.2.4.1. **(Added-AFMC)** AFMC Metal Fabrication & Technologies. Aircraft structural maintenance, metal fabrication and technologies also include forming, cutting, bending, and fastener replacement.

1.3.2.5. Nondestructive inspection is an inspection process or technique designed to reveal the damage at or beneath the external surface of a part or material without adversely affecting the material or part being inspected. Nondestructive inspection methods require specific long term training, experience and qualifications. AFMC Nondestructive Inspection experts provide engineering and technical support to program managers and MAJCOMs on nondestructive inspection standards, methods, processes, equipment, and personnel certification.

1.3.2.5.1. **(Added-AFMC)** Nondestructive inspection may detect flaws, measure geometric characteristics, determine material structure or composition, or it may characterize physical, electrical, or thermal properties without causing changes in the part impairing its future usefulness. The five primary methods of NDI are liquid penetrant, magnetic particle, eddy current, ultrasonic, and radiography. Nondestructive inspection methods require specific long term training, experience and qualifications. Visual inspections do not require formal NDI training and qualification and are not covered by this instruction.

1.4. **Waivers and Tailoring for Acquisition Programs.**

1.4.1. Waivers. Waivers for compliance requirements that this instruction places on the acquisition execution chain are not elevated through the MAJCOM-wing-unit organizational chain of authority. The acquisition execution chain is defined in AFI 63-101/20-101, *Integrated Life Cycle Management*. It includes the Service Acquisition Executive, Program Executive Officer, Decision Authority, the program manager, or other program office members. Therefore, tiering in accordance with the standard organizational terminology of AFI 33-360 cannot be applied. Instead, the waiver authority for each compliance requirement applied to the acquisition execution chain by this instruction is specified in the text.

1.4.2. Tailoring. Acquisition program managers, with Decision Authority approval, retain the ability to tailor and streamline strategies, oversight, reviews, phases, decision levels, documentation, regulatory requirements and information consistent with the tailoring guidance in AFI 63-101/20-101 and DoDI 5000.02T. Tailoring is especially appropriate for programs that are rapidly fielding capabilities, for example, Middle Tier Acquisition pathway programs or the acquisition of commercial off-the-shelf systems.

2. Roles and Responsibilities.

2.1. **Assistant Secretary of the Air Force for Acquisition, Technology & Logistics (SAF/AQ):**

2.1.1. Oversees Air Force Aircraft Structural Integrity Program policy.

2.1.2. Supports aircraft program office Aircraft Structural Integrity Program implementation as a part of the integrated life cycle management of systems.

2.1.3. Works with the Deputy Chief of Staff, Logistics, Engineering and Force Protection (AF/A4) and the Commander, AFMC (AFMC/CC) to designate the Air Force Aircraft Structural Integrity Program Technical Advisor.

2.1.4. Works with AF/A4 and AFMC/CC to develop and implement technician and artisan certification and recertification standards applicable to structural maintenance.

2.2. **Deputy Chief of Staff, Logistics, Engineering and Force Protection (AF/A4):**

2.2.1. Integrates Aircraft Structural Integrity Program into field-level maintenance policy.

2.2.2. Supports Aircraft Structural Integrity Program-related planning, programming, and budgeting activities for each aircraft program.

2.2.3. Promotes initiatives, develops policy and issues implementation guidance that enhances effectiveness of structural materials and fabrication maintenance. AF/A4 also resolves training issues in accordance with AFI 36-2651, *Air Force Training Program,* for the structural disciplines that directly involve Air Force Specialty Code 2A7XX and the equivalent civilian workforce.

2.3. **Commanders, Major Commands (MAJCOM) and Director, Air National Guard:**

2.3.1. Establish an Aircraft Structural Integrity Program office of primary responsibility for managing the Aircraft Structural Integrity Program command-wide.

2.3.2. Develop, publish, and update documentation specifying MAJCOM or lead operating command Aircraft Structural Integrity Program responsibilities in accordance with this instruction.

2.3.3. Assist program managers in the development, maintenance, and implementation of Aircraft Structural Integrity Program master plans as requested.

2.3.4. Install, operate, and maintain Aircraft Structural Integrity Program hardware. This includes systems that support loads and environment spectra surveys and individual aircraft tracking data collection, transfer, and analysis. These systems are maintained in a manner sufficient to support the valid data capture rates required by the aircraft-specific Aircraft Structural Integrity Program Master Plans.

2.3.5. Ensure that unit personnel collect and report data in accordance with the valid data capture rates defined by the Aircraft Structural Integrity Program Master Plans.

2.3.6. Assist program office Aircraft Structural Integrity Program managers in the development and implementation of corrective actions as needed to achieve required valid data capture rates.

2.3.7. Evaluate annual Aircraft Structural Integrity Program summaries as they apply to force structure, aircraft operational use and budgeting.

2.3.8. Ensure that unit personnel conduct structural inspections required by an aircraft's Aircraft Structural Integrity Program in accordance with the appropriate schedules, processes, procedures, and TOs.

2.3.9. Support command-wide integrated life cycle execution of structural disciplines by:

2.3.9.1. Assigning MAJCOM functional manager(s) to each structural discipline as determined by and applicable to the specific MAJCOM.

2.3.9.2. Ensuring personnel performing nondestructive inspections (including contracted personnel) are certified in accordance with AFI 21-101 and National Aerospace Standard 410, *Certification & Qualification of Nondestructive Test Personnel,* as applicable. **Note:** The AFMC nondestructive inspection focal point may

periodically audit and validate the certification of contracted nondestructive inspection personnel.

2.3.9.3. Providing subject matter expert support for TO change validation and verification processes.

2.3.9.4. Supporting field testing of materials, processes, facilities, and equipment.

2.3.9.5. Supporting proficiency testing and probability of detection studies in operational and depot environments.

2.3.9.6. Participating, as applicable to the MAJCOM structural disciplines, in periodic MAJCOM-specific surveys, and base assessments.

2.3.9.7. **(Added-AFMC)** Participating, as applicable to weapon system-specific advisory boards, such as the Corrosion Prevention Advisory Board (CPAB), for each weapon system within their area of responsibility (AOR).

2.4. **Commander, AFMC (AFMC/CC):**

2.4.1. Includes Aircraft Structural Integrity Program master plan development, maintenance, and implementation requirements in MAJCOM self-assessment communicator checklists.

2.4.2. Sustains and enhances a data management capability to support Aircraft Structural Integrity Program (e.g., the Aircraft Structural Integrity Management Information System).

2.4.3. Provides structural management integrated life cycle engineering and technical support to Headquarters Air Force, MAJCOMs, program managers, and to Air Force advisory boards for design, test, redesign, analysis and sustainment of assets associated with structural disciplines. This engineering and technical support includes:

2.4.3.1. Evaluating materials, processes, facilities and equipment requirements for structural impacts. This includes viability assessment of emerging material, nondestructive inspection, and repair technologies in structural disciplines and assistance with transition to new and existing systems.

2.4.3.2. Supporting Air Force career field managers, MAJCOM functional managers, Air Education and Training Command training managers, utilization and training workshops, and career development course writers for education and training of Air Force Specialty Code 2A7XX personnel.

2.4.3.3. Conducting Air Force structural management assessments and corrosion surveys at least every five years or at the request of a MAJCOM. These assessments and surveys identify deficiencies in the Air Force implementation of structural management disciplines and assist in the prioritization of AFMC structural management projects. Execute these assessments and surveys in accordance with AFI 90-201, *The Air Force Inspection System*.

2.4.3.4. Coordinating with environment, safety, and occupational health offices of primary responsibility to assist users in meeting applicable environment, safety, and occupational health requirements. This includes providing inputs associated with structural disciplines to Air Force occupational safety and health publications.

2.4.3.5. Establishing AFMC offices and focal points that provide technical experts for the structural disciplines listed in **paragraph 1.3.2** and designating managers for the structural management-related TOs identified in **Attachment 2** to ensure their currency, applicability, and completeness.

2.4.3.6. Providing program managers and MAJCOMs verifiable engineering inspection reliability data based on probability of detection and proficiency studies. These studies are conducted in accordance with Military Handbook 1823 (MIL-HDBK-1823), *Nondestructive Evaluation System Reliability Assessment* and Military Standard 1530 (MIL-STD-1530), *DoD Standard Practice for Aircraft Structural Integrity Program,* every five years or sooner when requested by MAJCOMs or program managers.

2.4.3.7. Establishing and maintaining the Air Force-wide process for the certification of all Air Force civil service nondestructive inspection personnel who require certification in accordance with AFI 21-101 or this instruction.

2.4.3.8. **(Added-AFMC)** Conduct an Air Force Materiel Command Nondestructive Evaluation and Inspection Executive Steering Group (ESG) meeting on an annual basis. ESG Principal Members include the Directors of AFLCMC/EN-EZ, AFRL/RX, AFSC/EN and AFNWC/EN.

2.4.4. **(Added-AFMC)** Air Force Life Cycle Management Center (AFLCMC) shall: **(T-3)**

2.4.4.1. **(Added-AFMC)** Establish and maintain the Air Force Structural Management Offices (AFSMO) in the disciplines of advanced composites, corrosion prevention and control, metals technology (and aircraft structural maintenance, i.e., sheet metal), and nondestructive inspection within the AFLCMC Product Support Engineering Division (AFLCMC/EN-EZ). **(T-3)** The low observables supportability discipline is managed by the AFLCMC LO Center of Excellence (AFLCMC/EN-EZ).

2.4.4.1.1. **(Added-AFMC)** AFLCMC/EN-EZ shall plan, program, and submit budget requests to support AFSMO disciplines and ensure program requirements are met.

2.4.4.2. **(Added-AFMC)** Each AFSMOs shall perform the following responsibilities within their respective area of technical expertise:

2.4.4.2.1. **(Added-AFMC)** Manage technical orders, as identified in **Attachment 3**.

2.4.4.2.2. **(Added-AFMC)** Assist with transition of new technologies to new and existing weapon systems.

2.4.4.2.3. **(Added-AFMC)** Provide engineering assistance for systems supportability and maintainability to field and depot level maintenance personnel.

2.4.4.2.4. **(Added-AFMC)** Support the Program Offices in system acquisition reviews and milestone activities.

2.4.4.2.5. **(Added-AFMC)** Provide engineering assistance and technical consultation across the Department of the Air Force.

2.4.4.2.6. **(Added-AFMC)** Provide technical data and information to support training at all levels of maintenance to the maximum extent feasible and possible. Conduct assistance visits as requested by Air Education and Training Command (AETC) technical schools, Air Logistics Complex (ALC) training functions, and other training units or as deemed necessary by the relevant AFSMO.

2.4.4.2.7. **(Added-AFMC)** In addition, to the responsibilities identified above, AFSMOs shall provide the additional support/functions as separately defined in **paragraph 4** of this supplement.

2.4.4.2.8. **(Added-AFMC)** Establish field level task specific inspection requirements for new or modified field level inspection procedures.

2.4.4.2.9. **(Added-AFMC)** Notify the Program Managers (PMs)/Product Group Managers (PGMs) of changes to centrally procured equipment and appropriate changes to weapon system-specific and commodity TOs.

2.4.4.3. **(Added-AFMC)** Program Executive Officers (PEOs) will verify all Time Compliant Technical Orders (TCTOs) involving the Air Logistic Complexes (ALCs) and AFSC 2A7X2 workforce in accordance with EZ-SB-15-002, *Requirements for NDI Procedure Development, Validation, and Verification for Aircraft Structural Inspections During Operations and Support Phase*. This shall be a self-inspection checklist item. Coordinate TO changes with the ALC NDI Program Manager. Ensure representatives from the affected weapon system and product wings/directorates participate in NDI Teams, Integrated Product Teams (IPTs), and conferences as needed to support the resolution of field and depot related issues. The Single Program Managers (SPMs) and Product Group Managers (PGMs) are crucial to the success of the AF structural maintenance programs, for they manage every aspect of their respective weapon systems.

2.4.4.4. **(Added-AFMC)** Program Managers (PMs)/Product Group Managers (PGM) shall: (**Note:** Overall authority, accountability, responsibilities of PMs are IAW DOD 5000 series direction and AFI 63-101/20-101, Acquisition and Sustainment Life Cycle Management. **Note:** NDI shall be accomplished IAW TO 33B-1-1, NDI Methods, Basic Theory, and TO 33B-1-2, Nondestructive Inspection General Procedures and Process Controls. Nondestructive Inspections include methods requiring specific long term training (formal and OJT (on-the-job training)) and qualifications. Personnel performing NDI inspections must be certified IAW AFI 21-101, TO 33B-1-1 and/or NAS 410 as applicable.)

2.4.4.4.1. **(Added-AFMC)** Coordinate with and/or inform stakeholder MAJCOMs, e.g., AFMC, USSF/Space System Command (SSC), and Lead Command regarding system facilities, tooling, equipment, test, technical data and training requirements impacting the structural disciplines addressed in this publication.

2.4.4.4.2. **(Added-AFMC)** Ensure PM/PGM initiated contract support requirements for NDI personnel contain provisions to have certified personnel IAW NAS 410 and that the certification is validated by the Air Force Nondestructive Inspection Office (AFNDIO).

2.4.4.4.3. **(Added-AFMC)** Evaluate the impact of proficiency testing or Probability of Detection (POD) against EN-SB-08-012 and aircraft inspection requirements. Refer to MIL-STD-1530, *Aircraft Structural Integrity Program (ASIP)*. Maintain, update and apply EN-SB-08-012 to aircraft programs.

2.4.4.4.4. **(Added-AFMC)** Program Managers (or their delegate) shall establish a charter to govern the NDI Team (NDIT) in accordance with MIL-STD-1530. The NDIT shall have the authority and responsibility to evaluate and implement appropriate NDI processes into all phases of the program, and for resolving system-specific issues and action items. The single manager, serving as Chair, conducts the NDIT periodically.

2.4.4.4.5. **(Added-AFMC)** Advise stakeholder MAJCOMs, e.g., AFMC, USSF/SSC, and Lead Command, on results of Time Compliant Technical Orders (TCTOs) impacting structural management and affecting safety, status and disposition of safety-related AFTO Form 252s and 22s, and flaw data on safety structures.

2.4.4.4.6. **(Added-AFMC)** Establish, maintain, and update the system-specific NDI program requirements and NDI technical manuals for which they have engineering responsibility. **(T-3)**

2.4.4.4.7. **(Added-AFMC)** PM/PGM shall coordinate with the ALC NDI Manager to ensure all new or modified NDI procedures are verified and validated per EZ-SB-15-002.

2.4.4.4.8. **(Added-AFMC)** Inform MAJCOMs on all technical compliance issues and changes in procedures and requirements.

2.4.4.4.9. **(Added-AFMC)** Identify and submit new NDI field inspection requirements and equipment to the AFNDIO for incorporation into equipment commercial item descriptions.

2.4.4.4.10. **(Added-AFMC)** Establish and maintain weapon system specific Nondestructive Inspection requirements for facilities, tooling, and equipment, including master NDI reference standards. Identify centrally procured NDI weapon system equipment needs to the AFNDIO.

2.4.4.4.11. **(Added-AFMC)** Ensure specialized weapon system NDI training is provided to maintenance personnel for unique NDI technology/equipment/procedures on their weapons systems.

2.4.4.4.12. **(Added-AFMC)** Ensure field NDI SMEs support and verify all new or modified field-level NDI procedures before they are published and distributed. Likewise, ensure depot maintenance SMEs support depot-only procedures.

2.4.4.4.13. **(Added-AFMC)** Conduct and lead root cause analysis investigations and implement corrective actions for inspection misses that impact aircraft safety or availability in accordance with Technical Order 33B-1-1 Appendix C, "*NDI Miss, Root Cause Analysis Guidelines."*

2.4.4.4.14. **(Added-AFMC)** Provide to the AFNDIO, all currently collected cost and manpower investment data needed to assess total cost of NDI inspections for

their weapon system.

2.4.4.4.15. **(Added-AFMC)** Exchange information about nondestructive inspections with the NDI community, specifically the AF NDI maintenance operations supporting their weapon system, on official weapon systems / program office websites, SharePoint or Teams page (as appropriate), to include:

2.4.4.4.15.1. **(Added-AFMC)** Results of TCTOs on safety-of-flight nondestructive inspections.

2.4.4.4.15.2. **(Added-AFMC)** Status and disposition of all NDI related AFTO Form 242s and 22s.

2.4.4.4.15.3. **(Added-AFMC)** Flaw data on safety-of-flight structures.

2.4.4.4.15.4. **(Added-AFMC)** Results from NDIT meetings.

2.4.4.4.15.5. **(Added-AFMC)** Results from MIL-STD-1530, *Aircraft Structural Integrity Program*, directed NDI Requirements Review Boards.

2.4.4.5. **(Added-AFMC)** The Support Equipment & Vehicle Division (SE&V, AFLCMC/WNZ) is the Air Force Office with Operational Support, Sustainability and Engineering (OSS&E) authority for ground support equipment with the exception of Non-Destructive Inspection (NDI) equipment. SE&V will provide Program Management for common NDI equipment to include logistics, contract management, item management, production management, and cataloging. SE&V procures NDI equipment based on validated, known requirements. Engineering responsibility and support is provided by the AFNDIO to include providing specifications and technical assistance, as well as validating new NDI equipment either prior to procurement or after procurement via bid samples or first articles. SE&V shall:

2.4.4.5.1. **(Added-AFMC)** Assign OSS&E authority for NDI equipment to the AFNDIO.

2.4.4.5.2. **(Added-AFMC)** Assign Program/Logistics Manager to manage the NDI equipment sustainment strategy.

2.4.4.5.3. **(Added-AFMC)** Coordinate with the AFNDIO on all equipment technical issues.

2.4.4.5.4. **(Added-AFMC)** Review and coordinate on all existing equipment acquisition changes or new additional equipment proposed by AFNDIO.

2.4.4.5.5. **(Added-AFMC)** Provide sustainment support for NDI equipment used by the Air Force NDI laboratories.

2.4.4.5.6. **(Added-AFMC)** Provide NDI equipment technical order provisioning and support of new equipment acquisitions.

2.4.4.5.7. **(Added-AFMC)** Provide cybersecurity assessment and authorization support for information technology systems.

2.4.4.5.8. **(Added-AFMC)** Coordinate with AFNDIO on future years defense program (FYDP) funding projections, prioritizations and execution of funds for NDI equipment acquisition, maintenance contracts, cybersecurity and equipment

training.

2.4.5. **(Added-AFMC)** Air Force Research Laboratory (AFRL) shall: **(T-3)**

2.4.5.1. **(Added-AFMC)** Coordinate and maintain prioritized documentation of Science and Technology (S&T) requirements for the structural disciplines.

2.4.5.2. **(Added-AFMC)** Dedicate research and development investments for the structural disciplines.

2.4.5.3. **(Added-AFMC)** Maintain appropriate competencies, lab capabilities, and responsive technical support for each of the structural disciplines.

2.4.5.4. **(Added-AFMC)** Materials and Manufacturing Directorate (AFRL/RX). The Materials and Manufacturing Directorate develops new materials, processes and manufacturing technologies for use in aerospace applications. This includes aircraft, spacecraft, missiles, rockets and ground-based systems and their structural, electronic and optical components. The directorate also provides quick reaction support and real time solutions to Air Force weapon system acquisition offices, field organizations and maintenance depots to solve materials related concerns and problems. The directorate plans, executes and integrates advanced manufacturing technology programs and affordability initiatives addressing manufacturing process technologies, computer integrated manufacturing and excellence through design for military needs. The directorate is also responsible for the Air Force technology programs that address environmental issues and provides materials expertise for airbase assets such as runways and infrastructures and technologies for aerospace expeditionary forces. Establish policies for management and continuity of the AF NDI program. This includes all functions from basic research through field implementation.

2.4.5.4.1. **(Added-AFMC)** Materials Integrity Branch (AFRL/RXSA) shall provide Research Development Testing and Evaluation (RDT&E) rapid response services including laboratory support and technical consultation with timely, accurate solutions to high priority, time-urgent USAF materials and process needs, including (1) delivering expertise for materials and process issues to include support for selection and substitution in USAF systems engineering processes; (2) providing test and characterization methods to understand and characterize real-world material service life behavior with delivered technical data; (3) providing independent failure analysis and recommended process solutions; (4) rapidly prototyping repair/inspection kit solutions to include technical support for rapid implementation and training; and (5) documenting lessons learned and establishing Materials and Processes (M&P) best practices through specifications and standards.

2.4.5.4.2. **(Added-AFMC)** Materials State Awareness Branch (AFRL/RXCA) shall lead, discover, develop, and deliver nondestructive evaluation (NDE) and material state awareness technologies and capabilities to assure maximum reliability and availability of Air Force Systems, including legacy and future vehicles, propulsion systems, and related systems and components.

2.4.6. **(Added-AFMC)** Air Force Sustainment Center (AFSC) shall: **(T-3)**

2.4.6.1. **(Added-AFMC)** The AFSC Commander or Executive Director shall appoint the AFSC NDI Manager to execute the AFSC NDI program and formally notify the Air Force Nondestructive Inspection Office (AFNDIO, AFLCMC/EZPT-NDIO) via appointment letter.

2.4.6.1.1. **(Added-AFMC)** The AFSC NDI Manager shall report to the AFSC Director of Engineering and the AFNDIO on an annual basis as to the state of the NDI Program in AFSC.

2.4.6.1.2. **(Added-AFMC)** The AFSC NDI Manager shall accomplish assessments of 100% of Depot level NDI laboratories in coordination with AFNDIO, at least every five (5) years using an AFNDIO approved NDI lab assessment checklist. The AFNDIO will provide assistance when requested.

2.4.6.1.2.1. **(Added-AFMC)** AFSC NDI Manager shall identify discipline, housekeeping, and technical discrepancies, attempt to identify the underlying cause for the deficiencies and recommend corrective actions.

2.4.6.1.2.2. **(Added-AFMC)** Encompass all NDI flights of the Air Logistics Complexes.

2.4.6.1.2.3. **(Added-AFMC)** Produce objective reports and provide specific definitions of problem areas, appropriate directive references, and recommended corrective action following format and example of the AF NDI community assessment checklist as published on the official SharePoint site.

2.4.6.2. **(Added-AFMC)** Each Air Logistics Complex Commander or Executive Director shall appoint an ALC NDI Manager to execute the ALC NDI program and formally notify the AFSC NDI Manager and the AFNDIO via appointment letter.

2.4.6.2.1. **(Added-AFMC)** The Air Logistics Complex NDI Manager shall report to the ALC Technical Director and AFSC NDI Manager on an annual basis as to the state of the NDI Program at that ALC.

2.4.6.2.2. **(Added-AFMC)** Each ALC NDI Manager shall assist the AFSC NDI Manager in development, implementation, execution and maintaining a standardized Depot NDI Laboratory Assessment Program.

2.4.6.3. **(Added-AFMC)** ALCs will standardize NDI processes where possible and interface with the appropriate Level 3 and the AFNDIO to ensure field needs are met. Ensure only properly trained and certified personnel perform NDI-related activities per AFSCI 20-114 and AFMCI 21-100.

2.4.6.4. **(Added-AFMC)** Each ALC within AFSC will assign a corrosion manager that possesses training, education and work experience in corrosion prevention and control to manage the ALC corrosion program which includes:

2.4.6.4.1. **(Added-AFMC)** Review changes to weapon-system-specific corrosion prevention and control technical data for impacts on depot operations.

2.4.6.4.2. **(Added-AFMC)** Review and provide technical advice for corrosion prevention and control aspects of corrosion control facilities, equipment, materials, and processes used at the ALC.

2.4.6.4.3. **(Added-AFMC)** Attend the Aircraft Structural Integrity Programs (ASIP) reviews conducted at their respective ALC when practicable.

2.4.6.5. **(Added-AFMC)** The AFSC NDI Manager and the ALC NDI Managers shall:

2.4.6.5.1. **(Added-AFMC)** Possess a National Aerospace Standard Certification & Qualification of Nondestructive Test Personnel (NAS 410) Level 3 certification in at least three (3) methods. For methods where the AFSC and/or ALC NDI Manager does not currently have a Level 3 certification or requires additional support, Level 3 personnel shall be designated to provide this support.

2.4.6.5.2. **(Added-AFMC)** Ensure all new or modified NDI procedures are formally validated and verified before TO change publication and distribution IAW AF guidance, including TO 33B-1-1, TO 33B-1-2, EZ-SB-15-002, MIL-HDBK-1823 and MIL-STD-1530. Exceptions to performance verification must be coordinated and approved by the AFSC and/or ALC NDI Manager and the responsible Weapon System Chief Engineer or delegate.

2.4.6.5.3. **(Added-AFMC)** Review and coordinate on all NDI procedures/changes submitted IAW published guidelines (including EZ-SB-15-002).

2.4.6.5.4. **(Added-AFMC)** Assist Program Managers (PMs)/Product Group Managers (PGMs) with securing field-level Subject Matter Expert (SME) support for TO change validation/ verification processes in conjunction with the appropriate MAJCOM functional manager.

2.4.6.5.5. **(Added-AFMC)** Maintain qualification/certification program. Review and approve training and proficiency requirements, including task specific training, for ALC NDI personnel. Refer to AFSCI 20-114, Qualification of Nondestructive Inspection Personnel. Function as Responsible Level 3 per NAS 410.

2.4.6.5.6. **(Added-AFMC)** Establish criteria to determine which complex inspections require additional media based training aids.

2.4.6.5.7. **(Added-AFMC)** Ensure that NDI personnel are sufficiently trained on new equipment and procedures before implementation.

2.4.6.5.8. **(Added-AFMC)** Determine if AETC support is required.

2.4.6.5.9. **(Added-AFMC)** Coordinate with Weapon System Chief Engineer or delegate to ensure specialized weapon system/item NDI training is provided to maintenance personnel for unique NDI procedures on their weapons systems.

2.4.6.5.10. **(Added-AFMC)** Ensure an AF NAS 410 Level 3 review and coordinate on all changes to NDI facilities, equipment, and processes in their respective organization when they impact inspection capability, sensitivity, or change parameters.

2.4.6.5.11. **(Added-AFMC)** Plan, program, and submit budget requests to support the resolution of depot related issues.

2.4.6.5.12. **(Added-AFMC)** Plan, program, and submit budget request to participate in Weapon System NDI Teams, NDI Advisory Boards, NDI IPTs, the

AF NDI Working Group, the AF NDI Managers Executive Working Group and the AF NDE/NDI Executive Steering Group meetings as needed to support the resolution of field and depot related issues. Encourage representatives from the affected weapon system and product directorate to participate.

2.4.6.5.13. **(Added-AFMC)** When authorized or delegated by the PM/PGM, the ALC NDI Manager shall oversee the purchase of specialized NDI equipment.

2.4.6.5.14. **(Added-AFMC)** Ensure only properly trained/certified personnel perform NDI-related activities. **(T-3)**

2.4.6.6. **(Added-AFMC)** 404 Supply Chain Management Squadron (SCMS) shall:

2.4.6.6.1. **(Added-AFMC)** Assign a manager for Allowance Standard (AS) 455 NDI and conduct a full review of AS 455 NDI at least every five (5) years. The AFNDIO will co-chair the review. Attendees will include MAJCOM functional managers, Equipment Managers, and Subject Matter Experts, and representatives from the AF NDI Schoolhouse, ALCs, and AFRL. Provide information copies of any interim changes (e.g., TACRs) to the AFNDIO **(T-3)**.

2.5. **Air Force Aircraft Structural Integrity Program Technical Advisor:**

2.5.1. Reviews all Aircraft Structural Integrity Program master plans including updates and makes approval recommendations to the Program Executive Officer. **(T-2).**

2.5.2. Reviews execution of the Aircraft Structural Integrity Program for each mission design series. **(T-2)**. Identifies trends, gaps, and opportunities for improvement in the Aircraft Structural Integrity Program execution, and will report these findings to Program Executive Officers and lead and operating MAJCOMs. **(T-2)**.

2.6. **Program Executive Officers (or delegated Decision Authority, when delegated in accordance with AFI 63-101/20-101):**

2.6.1. Ensure that program managers establish Aircraft Structural Integrity Programs and develop Aircraft Structural Integrity Program Master Plans for each mission design series operated by the Air Force in their portfolio.

2.6.2. Ensure aircraft program managers are executing the Aircraft Structural Integrity Program requirements in a timely manner as an integral part of life cycle management.

2.6.3. Utilize Aircraft Structural Integrity Program information in modification management and implementation.

2.6.4. Approve all Aircraft Structural Integrity Program master plans including updates.

2.7. **Program Managers:**

2.7.1. Plan, program, and budget funds required to develop, maintain, and execute the Aircraft Structural Integrity Program.

2.7.2. Ensure that program life cycle cost estimates developed in accordance with AFI 65-508, *Cost Analysis Guidance and Procedures,* include estimated costs of supporting the Aircraft Structural Integrity Program.

2.7.3. Oversee the development, modification and execution of the Aircraft Structural Integrity Program and ensure that the Aircraft Structural Integrity Program Master Plan is approved, updated and executed.

2.7.4. Unless waived in writing by the Program Executive Officer, will appoint an Aircraft Structural Integrity Program manager.

2.7.5. Ensure the usage collection and evaluation systems for loads and environment spectra survey and individual aircraft tracking achieve the valid data capture rates defined in the approved Aircraft Structural Integrity Program Master Plan. Monitor all Aircraft Structural Integrity Program flight data retrieval results to identify incomplete or missing data. Notify MAJCOM Aircraft Structural Integrity Program offices of primary responsibility of flight data retrieval discrepancies.

2.7.6. Ensure processes are in place for collecting, processing, storing, analyzing and reporting structural maintenance data essential to evaluating aircraft structural integrity.

2.7.7. Unless waived in writing by the Program Executive Officer, will annually provide summaries to lead commands of past aircraft operational usage and the effect on the structural integrity.

2.7.8. Establish and document the life cycle inspection and modification actions and schedules required to maintain the structural integrity of each aircraft system.

2.7.9. Ensure Aircraft Structural Integrity Program aircraft usage data requirements are integrated with mandatory crash survivable data recorder requirements whenever possible.

2.7.10. Coordinate new or modified nondestructive inspection procedures with the center nondestructive inspection manager or designated individual, who is Level 3 certified in accordance with National Aerospace Standard 410. **Note:** Level 3 certification ensures that the individual is capable of developing, qualifying and approving nondestructive test and inspection procedures, methods, and techniques.

2.7.11. Ensure contract support requirements for nondestructive inspection personnel contain provisions to have certified personnel in accordance with National Aerospace Standard 410.

3. Aircraft Structural Integrity Program Requirements and Procedures.

3.1. Unless waived in writing by the Decision Authority, the program manager shall establish an Aircraft Structural Integrity Program for each mission design series the Air Force acquires, uses, or leases.

3.2. For each aircraft mission design series developed or modified by the Air Force, the program manager shall implement an Aircraft Structural Integrity Program that complies with MIL-STD-1530, unless waived in writing by the Decision Authority. For these mission design series, unless waived in writing by the Decision Authority, the program manager shall:

3.2.1. Draft an initial Aircraft Structural Integrity Program Master Plan for the program as early as possible in the Technology Maturation and Risk Reduction phase. The initial Aircraft Structural Integrity Program Master Plan shall identify the tasks required to achieve structural integrity and to determine structural safety, performance, durability, supportability, and life cycle costs for the aircraft structure.

3.2.2. Obtain Program Executive Officer (or delegated Decision Authority) approval for the Aircraft Structural Integrity Program Master Plan before the system requirements review.

3.2.3. Update the Aircraft Structural Integrity Program Master Plan during the Engineering and Manufacturing Development, Production and Deployment, and Operations and Sustainment phases of the program to document changes in the Aircraft Structural Integrity Program.

3.2.4. For aircraft in sustainment, execute the Aircraft Structural Integrity Program as an integral part of the total system engineering and management effort in the sustainment of the aircraft.

3.2.5. For aircraft that are to be modified, fly new missions, or whose operation will extend past the aircraft's certified design service life, develop a revised Aircraft Structural Integrity Program Master Plan. Obtain Program Executive Officer approval of the revised plan before modifications are executed, regular flights begin under the new mission, or commencing operations beyond the previously certified service life.

3.3. For each aircraft mission design series operated by the Air Force but not developed or modified by the Air Force, the program manager shall, unless waived in writing by the Program Executive Officer, use Military Standard 1530 as the basis for implementing Aircraft Structural Integrity Program tasks and elements in the program. Program managers select and implement tasks and elements necessary to ensure the aircraft's structural safety, performance, durability, supportability, and affordability for the operational life of structural components, while remaining consistent with the program's acquisition strategy and engineering authority over the aircraft. For these mission design series, the program manager, unless waived in writing by the Program Executive Officer, shall:

3.3.1. Document the tailored program in an Aircraft Structural Integrity Program Master Plan.

3.3.2. Finalize the Aircraft Structural Integrity Program Master Plan and obtain Program Executive Officer (or delegated Decision Authority) approval before the Air Force operates the aircraft.

4. (Added-AFMC) AFMC Requirements as Relating to Structural Disciplines.

4.1. **(Added-AFMC)** Commander, AFMC (AFMC/CC) in addition to requirements in **paragraph 2.4** shall:

4.1.1. **(Added-AFMC)** Provide AF enterprise structural management integrated life cycle engineering and technical support to MAJCOMs, PMs/PGMs, other AF offices and AF advisory boards for design, test, redesign, analysis and sustainment of assets associated with structural disciplines. This includes:

4.1.1.1. **(Added-AFMC)** Material/proficiency testing processes and studies; development of methods and equipment; and life cycle management planning.

4.1.1.2. **(Added-AFMC)** Development, implementation, and standardization of procedures and processes to enhance structural materials fabrication and maintenance effectiveness:

4.1.1.2.1. **(Added-AFMC)** Coordinate with/obtain concurrence from USSF/SSC on procedures and process that apply to Space Programs.

4.1.1.2.2. **(Added-AFMC)** Support development and/or revision of standards, procedures and processes, and advice in the acquisition, processing and interpretation of data.

4.1.1.2.3. **(Added-AFMC)** Ensure currency, applicability, completeness, and designation of AF managers for the TOs listed at **Attachment 2**.

4.1.1.2.4. **(Added-AFMC)** Provide safety-related inputs associated with structural disciplines to Air Force Occupational Safety Fire and Health Standards, DAFMAN 91-203.

4.1.1.3. **(Added-AFMC)** Evaluation, as requested by MAJCOMs and PMs/PGMs, of materials, processes, facilities and equipment requirements for structural impacts.

4.1.1.4. **(Added-AFMC)** Viability assessment of emerging material and repair technologies in structural disciplines and assistance with transition to new and existing systems, to include participation in field-tests as requested by PMs/PGMs on material and process technologies.

4.1.2. **(Added-AFMC)** Coordinate with ESOH offices of primary responsibility (OPRs) to assist users in meeting applicable ESOH requirements.

4.1.3. **(Added-AFMC)** Conduct command-specific structural management surveys and assessments of representative number of activities, statistically derived by the respective AFMC structural management activity; coordination and scheduling being consistent with AFI 90-201, *The Air Force Inspection System*.

4.1.4. **(Added-AFMC)** Promote standardization and/or interoperability of equipment, data systems and software programs associated with structural-related disciplines.

4.1.5. **(Added-AFMC)** Support Air Force Career Field Managers (AFCFMs), MAJCOM Functional Managers, Air Education and Training Command (AETC) Training Managers, Specialty Training Requirements Team (STRT)/Utilization and Training Workshops (U&TWs), and Career Development Course (CDC) writers for education and training of Air Force Specialty Codes (AFSCs) 2A7XX personnel.

4.1.6. **(Added-AFMC) Advanced Composites.** Air Force Advanced Composites Office (ACO), assigned to AFLCMC/EN-EZ, shall serve as the focal point for the AF advanced composites integration and maintenance. The ACO shall provide technical leadership to guide the integration and maintenance support program for HQ USAF by overseeing composites-related maintenance and repair activities throughout the Air Force and provide technical leadership and support to users. The ACO shall:

4.1.6.1. **(Added-AFMC)** Plan, program, and budget for:

4.1.6.1.1. **(Added-AFMC)** Attendance at Air Force composite and advanced materials sustainment meetings.

4.1.6.1.2. **(Added-AFMC)** Development and maintenance of general series TO 1-1-690 *General Advanced Composite Repair Processes Manual*.

4.1.6.1.3. **(Added-AFMC)** Execution of advanced composite field assessments.

4.1.6.2. **(Added-AFMC)** Assist Program Managers/Product Group Managers (PMs/PGMs), Aircraft Battle Damage Repair (ABDR) Office, and MAJCOMs in providing engineering support in sustainment, processing, equipment and deployable rapid-repair capability for advanced composite structures.

4.1.6.3. **(Added-AFMC)** Serve as the Air Force (AF) focal point for AF advanced composite assistance visits.

4.1.6.3.1. **(Added-AFMC)** Conduct, upon request of MAJCOMs, field maintenance activities, or as deemed necessary by the ACO, assistance visits to identify and resolve aircraft repair-related issues.

4.1.6.3.2. **(Added-AFMC)** Report assistance visit results to Commanders, Inspector General (IG), ASIP Managers and MAJCOM representatives. Elevate concerns and recommended actions to leadership at the appropriate level.

4.1.6.4. **(Added-AFMC)** Manage Technical Order 1-1-690, *General Advanced Composite Repair Processes Manual* to ensure the accuracy, currency and timely publication.

4.1.6.5. **(Added-AFMC)** Provide engineering assistance and technical consultation to MAJCOMs, AFSC, ALCs, AFRL, Single Program Managers (SPMs), Product Group Managers (PGMs), and field maintenance personnel.

4.1.6.5.1. **(Added-AFMC)** As requested by MAJCOMs or as deemed necessary by the ACO, provide onsite training to AF engineers and advanced composite field maintenance personnel.

4.1.6.6. **(Added-AFMC)** Ensure minimum facility, tooling and equipment guidelines for generic advanced composites supportability processes are IAW TO 1-1-690, *General Advanced Composite Repair Processes Manual*.

4.1.6.7. **(Added-AFMC)** Host an annual composite and advanced materials sustainment meeting to identify, discuss and resolve composite issues. Include representation from MAJCOMs, AF Schoolhouse, AFMC Centers, Program Executive Officers (PEOs), Program Managers (PMs), Product Group Managers (PGMs), and sister services.

4.1.6.8. **(Added-AFMC)** Support composite additive manufacturing (AM, 3D printing).

4.1.6.9. **(Added-AFMC)** Provide Subject Matter Expert (SME) support to ensure technical accuracy, validity, and currency of crashed, damaged, or disabled aircraft recovery composite related issues.

4.1.6.10. **(Added-AFMC)** Maintain appropriate laboratory capabilities to support advanced composite processing, tooling, equipment engineering and field support operations.

4.1.6.11. **(Added-AFMC)** Represent the Air Force in committee and subcommittee meetings of national and international advanced composite standards organizations.

4.1.7. **(Added-AFMC)** Corrosion Prevention and Control. AF Corrosion Prevention and Control Office (AFCPCO), assigned to AFLCMC/EN-EZ, shall serve as the focal point for the AF corrosion prevention, mitigation and control. The AFCPCO shall provide technical leadership to guide the maintenance support program for AF air and space systems, by overseeing corrosion prevention, mitigation, and control, and corrosion-related maintenance and repair activities throughout the AF and provide technical leadership and support to users. The AFCPCO shall:

4.1.7.1. **(Added-AFMC)** Provide support to SAF/AQRE and the AF Corrosion Control and Prevention Executive (CCPE) therein by:

4.1.7.1.1. **(Added-AFMC)** Upon request, provide to the CCPE an annual report detailing health of air and space assets' corrosion prevention and control within the AF, to include trend analysis gathered from surveys and results from Corrosion Prevention Advisory Boards (CPABs).

4.1.7.1.2. **(Added-AFMC)** Support the AF Corrosion Prevention Advisory Board (AFCPAB).

4.1.7.1.3. **(Added-AFMC)** Monitor Cost of Corrosion for maintenance of USAF air and space assets.

4.1.7.1.3.1. **(Added-AFMC)** Participate in the USAF Corrosion Metrics Working-level Integrated Product Team (WIPT).

4.1.7.1.3.2. **(Added-AFMC)** Provide input to CCPE contracted party to determine inputs to the cost of corrosion algorithm.

4.1.7.1.4. **(Added-AFMC)** Participate in acquisition program reviews of new Air Force Systems and Subsystems to identify and mitigate corrosion concerns throughout system life cycle, and advise program managers of current corrosion prevention methods during design, prototyping, and initial fielding of new systems.

4.1.7.1.5. **(Added-AFMC)** Provide support in developing strategic roadmaps.

4.1.7.2. **(Added-AFMC)** Review and validate master facilities requirements for air and space assets' corrosion maintenance operations at all levels.

4.1.7.2.1. **(Added-AFMC)** Coordinate such requirements with Air Force Civil Engineering Center (AFCEC).

4.1.7.3. **(Added-AFMC)** Survey representative MAJCOM air and space assets' corrosion prevention and control operations/activities at the request of a MAJCOM functional manager or at least every 5 years. Pre-survey planning shall be coordinated with the MAJCOM Functional Manager, receiving the visit, who will interface with the MAJCOM Gatekeeper to establish the schedule; refer to AFI 90-201 for additional detail.

4.1.7.3.1. **(Added-AFMC)** Conduct weapon system (Model Design Series) specific survey and Staff Assistance Visit (SAV) upon request of the cognizant Program Office.

4.1.7.4. **(Added-AFMC)** Perform assistance to address:

4.1.7.4.1. **(Added-AFMC)** Specific field, Program Managers/Product Group Managers (PMs/PGMs), or MAJCOM concerns.

4.1.7.4.2. **(Added-AFMC)** Field tests, command/agency surveys, resolution of materials-related difficulties, and in-service equipment problems, when requested.

4.1.7.5. **(Added-AFMC)** Participate in corrosion materials and process technology exchange forums and joint sustainment activities within the AF, other services, DoD, Government agencies and industry.

4.1.7.5.1. **(Added-AFMC)** Conduct and host an annual Corrosion Technical Interchange Meeting (CTIM).

4.1.7.5.2. **(Added-AFMC)** Co-chair, in coordination with Naval Air Systems Command (NAVAIR) and US Army Aviation and Missile Command (AMCOM), the Office of the Secretary of Defense (OSD) Aerospace Systems Corrosion Control Working Group (ASCCWG) to discuss common:

4.1.7.5.3. **(Added-AFMC)** Past, present, and future corrosion prevention and mitigation efforts.

4.1.7.5.4. **(Added-AFMC)** Relevant policy changes and its effect on the acquisition of corrosion prevention and control (CPC) technologies, sustainment, and other subsidiary topics.

4.1.7.6. **(Added-AFMC)** Serve as the central information warehouse to develop and maintain best engineering practices, and qualified products lists (QPLs), to include equipment and materials as applicable in the general series technical orders.

4.1.7.6.1. **(Added-AFMC)** Develop and publish pertinent information and promotional materials through messages and a website to facilitate proficiencies of corrosion prevention and control maintenance activities.

4.1.7.7. **(Added-AFMC)** Support research and development (R&D) when requested.

4.1.7.7.1. **(Added-AFMC)** Provide test planning, field service evaluation partners, and environmental exposure testing sites for technology transition.

4.1.7.8. **(Added-AFMC)** Serve as AF liaison to organizations that maintain coating specifications and standards utilized by the AF and other sister services.

4.1.7.9. **(Added-AFMC)** Support development of:

4.1.7.9.1. **(Added-AFMC)** Procurement vehicles associated with coatings-related materials and equipment to include qualification testing and the required laboratory certifications needed for qualification testing, specification development and maintenance, and technical order support.

4.1.7.9.2. **(Added-AFMC)** Test and evaluation criteria to define the performance of the organic coating systems.

4.1.7.9.3. **(Added-AFMC)** Characteristics of coating systems for defining the window of application, removal and repair.

4.1.7.9.4. **(Added-AFMC)** Integration testing and scale-up for defining the

processes that marry the coating system to the application and removal equipment.

4.1.7.9.5. **(Added-AFMC)** Improved laboratory and field level test methods.

4.1.7.10. **(Added-AFMC)** Support Air Logistics Complexes and weapon system corrosion managers during the review of new and major revisions to technical data as requested by the appropriate manager.

4.1.7.11. **(Added-AFMC)** Provide info and assistance with respect to corrosion prevention and control (CPC) training in maintenance, program management, and engineering as well as the input to training course curricula as developed by AETC, Defense Acquisition University (DAU), or other responsible parties.

4.1.7.12. **(Added-AFMC)** Provide cognizant engineering support for: T.O. 1-1-8, *Application and Removal of Organic Coatings, Air and Space and Non-Air and Space Equipment*; T.O. 1-1-689, *Avionics Cleaning and Corrosion Prevention/Control*, T.O. 1-1-691, *Aircraft Weapon Systems Cleaning and Corrosion Control*; T.O. 35-1-3, *Corrosion Prevention, Painting and Marking of USAF Support Equipment*; and T.O. 1-1-700, Corrosion Prevention And Control For Ground Communication, Electronics And Meteorological (C-E-M) Equipment And Its Protective Shelters, Vans, and Radomes.

4.1.7.13. **(Added-AFMC)** Participate in and provide technical and engineering support to all AF weapon system Corrosion Prevention Advisory Boards (CPABs).

4.1.7.13.1. **(Added-AFMC)** Assist weapon system corrosion managers, MAJCOM functional managers, and Air Logistics Complexes (ALC) corrosion managers in tracking, researching, and resolving action items.

4.1.7.13.2. **(Added-AFMC)** Develop and maintain standardized guidelines for weapon system-specific CPAB charters. These guidelines will address constituency, roles, procedures, and responsibilities.

4.1.7.13.3. **(Added-AFMC)** As requested, help host face-to-face CPAB meetings annually, with invitations for participation extended to Original Equipment Manufacturer (OEM), field, MAJCOM, and AFCPCO representatives.

4.1.7.13.4. **(Added-AFMC)** Develop and maintain standardized guidelines for weapon system-specific corrosion prevention and control plans (CPCPs) IAW AFI 63-101/20-101 and the CPCP template published by the AF Corrosion Prevention and Control Office (AFCPCO). CPCPs shall be reviewed at least every two (2) years. **(T-3)** The CPCP and updates shall be coordinated through the AFCPCO prior to being published.

4.1.7.13.5. **(Added-AFMC)** Review CPCPs from Weapon System Program Offices upon request from the cognizant Program Office manager or at least once every two years.

4.1.7.14. **(Added-AFMC)** Participate in annual program management reviews with AF/A4M, AFLCMC/EZP, and the MAJCOM functional managers to assess and enhance program effectiveness for 2A7X3 and 2A7X5 AFSC career fields and civilian equivalents.

4.1.7.15. **(Added-AFMC)** Identify and evaluate new CPC sustainment technologies for transition to the warfighter to improve the service life, reduce the cost of maintenance, and improve weapon system availability.

4.1.8. **(Added-AFMC) Low Observables Supportability.** Low Observables (LO) Supportability, assigned to AFLCMC LO Center of Excellence (AFLCMC/EN-EZ), shall serve as the focal point for the integrity, reliability and maintainability of AF LO materials, repair processes, nondestructive inspections, signature diagnostic measurements, and signature health assessments. AFLCMC LO Center of Excellence shall:

4.1.8.1. **(Added-AFMC)** Support the AF Low Observables Integrity Program (LOIP). Manage MIL-HDBK-513, Air Vehicle Low Observable Integrity Program (LOIP) General Guidelines.

4.1.8.1.1. **(Added-AFMC)** Sponsor AFRL co-locates in an effort to increase organic engineering capability and ease transition of LO technology to fielded and future platforms.

4.1.8.1.2. **(Added-AFMC)** Assist AFRL/XP in conducting, consolidating and reporting the biennial (every other year) LO survey to include site visits.

4.1.8.1.3. **(Added-AFMC)** Coordinate and establish processes to notify stakeholders of changes to LO facilities, equipment, materials and processes at the weapon system level when they impact LO integrity, survivability, and maintainability, or when they impact common LO programs and solutions.

4.1.8.1.4. **(Added-AFMC)** Plan and participate in LO summits, advisory boards, Independent Review Teams (IRT), and Aircraft Low Observable Working Groups.

4.1.8.1.5. **(Added-AFMC)** Establish a working level group with Program Office engineers, maintainers, and Program Managers, and appropriate representatives from AFRL, AFSC, and MAJCOMs. The LO Working Group shall interact with operational units to understand, appreciate and influence technology decisions. Field experienced maintainers are essential in providing real world and recent experiences to this group.

4.1.8.2. **(Added-AFMC)** Assist/coordinate with the appropriate offices on changes to LO facilities, equipment, testing, materials and processes at the system level when they impact LO integrity, reliability and maintainability or when they impact common LO assets.

4.1.8.3. **(Added-AFMC)** Organize and/or participate in AF LO executive councils, advisory boards, independent review teams and aircraft LO working groups, and serve as the AF focal point for inter-service LO integrity and supportability-related meetings.

4.1.8.4. **(Added-AFMC)** Survey representative activities at least every five years or at the request of a MAJCOM for LO issues/concerns to assist in prioritizing future LO projects. Pre-survey planning shall be coordinated with the MAJCOM Functional Manager, receiving the visit, who will interface with the MAJCOM Gatekeeper to establish the schedule, refer to AFI 90-201 for additional detail.

4.1.9. **(Added-AFMC) Metal Fabrication & Technologies.** AF Metal Fabrication & Technologies Office (MFTO), assigned to AFLCMC/EN-EZ, shall serve as the focal point

for the Aircraft Metals Technology (MT) career field, Air Force Specialty Code (AFSC) 2A7X1, and the Aircraft Structural Maintenance (SM) career field, Air Force Specialty Code 2A7X3, in metal fabrication and metal fabrication related activities (related to non-corrosion prevention and control). The AF MFTO shall provide technical leadership to guide the maintenance support program for HQ USAF by overseeing MT and SM activities (facilities, materials, metal forming, bending, and cutting, fastener replacement, machining, welding, heat treating, and additive manufacturing), throughout the AF and provide engineering and technical support to users, program managers and Program Offices. The AF MTO shall:

4.1.9.1. **(Added-AFMC)** Establish standard equipment requirements with the assistance/ coordination of Program Managers/Product Group Managers (PMs/PGMs) and MAJCOMs.

4.1.9.1.1. **(Added-AFMC)** Provide engineering and technical support for procurement, repair, and maintainability of centrally procured manufacturing and metal fabrication equipment, and serve as the engineering authority for centrally procured MT and SM equipment, product and item specifications, and MT and SM specific equipment requirements, options, and upgrades.

4.1.9.1.2. **(Added-AFMC)** Evaluate new equipment to ensure adequate testing is accomplished prior to its fielding and to ensure equipment meets AF requirements for safety, deployability, sensitivity, repeatability, reliability, and maintainability.

4.1.9.1.3. **(Added-AFMC)** Develop and maintain standard equipment and machinery requirements list for field MT and SM sections.

4.1.9.1.4. **(Added-AFMC)** Provide technical support for Additive Manufactured equipment purchases in accordance with TO 34A-1-1.

4.1.9.2. **(Added-AFMC)** Provide and maintain appropriate guidance for welding certification.

4.1.9.2.1. **(Added-AFMC)** Maintain MAJCOM specific welding certification group requirements as established by the applicable technical orders.

4.1.9.2.2. **(Added-AFMC)** Provide consultation to the AF NDI Office and field units on weld certification specimen inspection requirements.

4.1.9.2.3. **(Added-AFMC)** Provide consultation to field level flight chiefs on Aerospace Ground Equipment (AGE) weld repairs, as requested.

4.1.9.3. **(Added-AFMC)** Support and implement equipment standardization to ease training, maintenance, manufacturing, fastener replacement, machining, welding, and heat treating operations and to improve operational effectiveness and safety across the enterprise.

4.1.9.3.1. **(Added-AFMC)** Budget, fund, and deliver enterprise Computer Aided Design/Computer Aided Manufacturing (CAD/CAM), and Solid Modeling software licenses, including software certification and software training. Engage with legal counsel to ensure computer software licenses are consistent with federal law and will meet user needs per the Air Force Federal Acquisition Regulation Supplement (AFFARS) 5301.602.

4.1.9.3.2. **(Added-AFMC)** Update cyber-security policy for use of off-network hardware (laptop, desktop computers) and software, specifically for aircraft maintenance, manufacturing and repair activities to enable innovative and advanced repair processes for aircraft and support equipment.

4.1.9.3.3. **(Added-AFMC)** Streamline and standardize MT Computer Numeric Control (CNC) equipment for field MT and SM sections.

4.1.9.4. **(Added-AFMC)** Works with AFLCMC Air Force Rapid Sustainment Office (AFLCMC/RSO), the focal point for Agile Manufacturing, additive manufacturing (AM), 3D printing, cold spray efforts and their implementation across the AF.

4.1.9.4.1. **(Added-AFMC)** Support manufacturing requests (forming, bending, milling, lathe, welding, heat treating, and AM) from PMs/PGMs and Program Offices. Coordinate with applicable MAJCOM Functional Managers to provide organic MT manufacturing/activity support.

4.1.9.5. **(Added-AFMC)** Survey representative MT and SM sections/activities at the request of the MAJCOM Functional Manager or at least every five (5) years. Pre-survey planning shall be coordinated with the MAJCOM Functional Manager, receiving the visit, who will interface with the MAJCOM Gatekeeper to establish the schedule, refer to AFI 90-201 for additional detail.

4.1.9.6. **(Added-AFMC)** Lead MT and SM specific assemblies or gatherings to address enterprise business matters internal to the Air Force, such as Product Improvement Working Groups (PIWG), or Executive Working Group (EWG), to solve MAJCOM unit and organizational issues.

4.1.9.7. **(Added-AFMC)** Establish and maintain a contract vehicle to support unique manufacturing equipment, software, and training requirements for all field MT and SM sections.

4.1.9.8. **(Added-AFMC)** Support, prepare, maintain, and disseminate notifications, policies, procedures, guidance and taskers in areas affecting field MT and SM sections to increase effectiveness or address safety and health related concerns, on an as-needed basis, or at the request of MAJCOMs.

4.1.10. **(Added-AFMC) Nondestructive Inspection (NDI).** The AF Nondestructive Inspection Office (AFNDIO), assigned to AFLCMC/EN-EZ, shall serve as the focal point for the AF NDI maintenance program. AFNDIO shall guide the maintenance support program for HQ USAF by overseeing NDI-related activities throughout the AF and provide technical support to users. The AFNDIO shall:

4.1.10.1. **(Added-AFMC)** Plan, program, and budget for:

4.1.10.1.1. **(Added-AFMC)** Equipment engineering and acquisition support and first article testing of initial issue NDI equipment.

4.1.10.1.2. **(Added-AFMC)** AFNDIO technology demonstrators and/or insertion requirements in coordination with the AF NDI Managers EWG to develop and maintain national, international and AF-level standards and practices for NDI.

4.1.10.1.2.1. **(Added-AFMC)** Attendance and participation in NDI Advisory

Boards.

4.1.10.1.3. **(Added-AFMC)** Development and maintenance of general NDI TOs (33B Series).

4.1.10.1.4. **(Added-AFMC)** Sponsor and host an AF-wide NDI Working Group (NDIWG), AF NDI Managers EWG, and attendance/participation in the AF NDE/NDI Executive Steering Group (ESG) meetings.

4.1.10.1.5. **(Added-AFMC)** Execution of the AFNDIO Assessment Program.

4.1.10.1.6. **(Added-AFMC)** Attendance and participation in Weapon System NDI Teams, NDI IPTs, and other technical events/meetings to ensure adequate representation of the AF NDI program.

4.1.10.2. **(Added-AFMC)** Provide engineering assistance and technical consultation to USAF/AFMM, MAJCOMs, AFSC, ALCs, AFRL, single managers (SPMs and PGMs) and field maintenance personnel.

4.1.10.2.1. **(Added-AFMC)** As requested by MAJCOMs, provide onsite training to AF NDI personnel.

4.1.10.2.2. **(Added-AFMC)** Manage the AF NDI Technical Community SharePoint and distribute career field technical information accordingly.

4.1.10.3. **(Added-AFMC)** Provide PMs/PGMs and MAJCOMs verifiable engineering inspection reliability data by conducting PoD and proficiency studies IAW MIL-HDBK-1823, *Nondestructive Evaluation System, and Reliability Assessment* and MIL-STD-1530, *Aircraft Structural Integrity Program (ASIP)*, in each method every five (5) years or sooner when requested by MAJCOMs or PMs/PGMs.

4.1.10.4. **(Added-AFMC)** Support the AFSC and ALC NDI Managers and Weapon System Chief Engineers during review of technical data revisions.

4.1.10.5. **(Added-AFMC)** Manage Technical Order 33B-1-1, *NDI Methods, Basic Theory;* and Technical Order 33B-1-2, *Nondestructive Inspection General Procedures and Process Controls* to ensure the accuracy, currency and timely publication.

4.1.10.6. **(Added-AFMC)** As the engineering authority assigned OSS&E for centrally managed and procured NDI equipment, provides engineering support for acquisition, repair and maintainability of NDI equipment to promote standardization and prevent and eliminate proliferation of NDI equipment.

4.1.10.7. **(Added-AFMC)** Establish NDI technical equipment requirements. Obtain inspection requirements from the Weapon System Chief Engineer, AFSC NDI Manager, ALC NDI Managers, AFRL/RX and MAJCOM functional managers to develop necessary equipment descriptions and specifications for central procurement.

4.1.10.8. **(Added-AFMC)** Evaluate new centrally managed equipment before procurement to ensure adequate field-testing is accomplished prior to its acquisition and to ensure equipment meets AF requirements for safety, deployability, sensitivity, repeatability, reliability and maintainability.

4.1.10.9. **(Added-AFMC)** Conduct an annual NDIWG to address enterprise NDI issues and ensure AF and inter-service standardization. Include representation from MAJCOMs, the AF NDI Schoolhouse 359 TRS DET 1/NDI, AFMC Centers PEO/PMs/PGMs and sister services.

4.1.10.10. **(Added-AFMC)** Conduct an Air Force NDI manager's Executive Working Group (EWG) meeting. The members of the EWG should include the AFNDIO, MAJCOMs, 359 TRS DET 1/NDI, AFSC and ALC NDI representatives, Structural Integrity Programs representatives, the NDI equipment Item Manager and Program Manager, AS 455 Manager, and AFRL R&D personnel.

4.1.10.11. **(Added-AFMC)** Participate in and provide technical and engineering support to all NDIT's. Assist weapon system single managers, MAJCOM functional managers, and ALC NDI managers in tracking, researching and resolving action items.

4.1.10.12. **(Added-AFMC)** Serve as the AF focal point for the AFNDIO Assessment Program.

4.1.10.12.1. **(Added-AFMC)** Maintain a method to assess NDI activities, including contractor operated AF-owned NDI laboratories, for compliance with TO 33B-1-1, DAFI 63-140, and DAFI 21-101.

4.1.10.12.2. **(Added-AFMC)** Establish and maintain a laboratory evaluation team of qualified senior non-commissioned officers who hold the AFSC 2A7X2 (NDI).

4.1.10.12.3. **(Added-AFMC)** The Evaluation Team shall:

4.1.10.12.3.1. **(Added-AFMC)** Assess the effectiveness and capability of organizational and intermediate level NDI laboratory. NDI laboratories shall be assessed IAW AFI 90-201, this AFI, and TO 33B-1-1 every 5 years or at the request of the MAJCOM functional manager, System Program Office, or Structural Integrity Community.

4.1.10.12.3.2. **(Added-AFMC)** The Evaluation Team Chief will approve or disapprove of any inspection augmentees from the field. Determinations will be made considering training, experience, and AFSC 2A7X2 Enlisted Development Team vectors.

4.1.10.12.3.3. **(Added-AFMC)** Report science and engineering based technical issues to the AF NDI EWG for discussion and elevation to the AF NDE/NDI ESG as appropriate.

4.1.10.12.3.4. **(Added-AFMC)** Report assessment results to Commanders, Inspector General (IG), ASIP Managers and MAJCOM representatives. Elevate concerns and recommended actions to leadership at the appropriate level.

4.1.10.12.3.5. **(Added-AFMC)** Administer the Proficiency Testing/Capability Assurance Program IAW TO 33B-1-1.

4.1.10.13. **(Added-AFMC)** Review and validate master facilities requirements for NDI operations.

4.1.10.14. **(Added-AFMC)** Support the Air Force NDI career field manager in establishing and maintaining the qualification and certification for AF NDI personnel.

4.1.10.14.1. **(Added-AFMC)** Air Force military personnel (Active Duty, Air National Guard, Reserves) are certified in accordance with procedures outlined in their Career Field Education and Training Plan (CFETP), AFI 21-101, and TO 33B-1-1 for Air Force Specialty Code (AFSC) 2A7X2. The process for qualification shall be coordinated with MAJCOM functional managers and Air Education and Training Command training managers, Utilization & Training Workshops (U&TW), CDC writers for education and training of AFSC 2A7X2 personnel.

4.1.10.14.2. **(Added-AFMC)** Establish and maintain the process for the qualification and certification of Field Level Civil Service AF NDI personnel requiring certification IAW the requirements of AFI 21-101, TO 33B-1-1, and NAS 410.

4.1.10.14.2.1. **(Added-AFMC)** As requested by MAJCOM units, provide onsite training or testing for certification or recertification of AF NDI personnel.

4.1.10.14.2.2. **(Added-AFMC)** When requested by the contracting officer / organization, ensure contracted NDI personnel are certified by their employer in accordance with the requirements of NAS 410. Compliance by the contractor with NAS 410 SHALL be audited and validated by the Air Force NDI Office AFLCMC/EZPT-NDIO at Tinker AFB, OK. Audits will occur at any time deemed necessary by the Air Force to ensure certification of contracted NDI personnel are in compliance with the requirements of NAS 410.

4.1.10.14.2.3. **(Added-AFMC)** Review third-parties providing certification services to contractor operated AF-owned NDI laboratories to ensure compliance with NAS 410, IAW TO 33B-1-1.

4.1.10.15. **(Added-AFMC)** Represent the AF in committee and subcommittee meetings of national and international NDI standards organizations.

4.1.10.16. **(Added-AFMC)** Distribute official information via the message system.

4.1.10.17. **(Added-AFMC)** Maintain appropriate laboratory capabilities to support equipment engineering and field support operations.

4.1.10.18. **(Added-AFMC)** Manage the Process Control Automated Maintenance System (PCAMS) IAW TO 33B-1-1.

William B. Roper, Jr.
Assistant Secretary of the Air Force
(Acquisition, Technology & Logistics)

(AFMC)

THOMAS M FISCHER, SES
Director, Engineering and Technical Management
USAF Technical Airworthiness Authority

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

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(Added-AFMC) AFI 20-114, *Air And Space Equipment Structural Management*, 2 April 2014

(Added-AFMC) AFMCI 21-100, *Depot Maintenance Management*, 6 September 2018

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AFI 21-101, *Aircraft and Equipment Maintenance Management,* 16 January 2020

(Added-AFMC) EN-SB-08-012, *In-Service Inspection Crack Size Assumptions for Metallic Structures*, 1 February 2018

(Added-AFMC) EZ-SB-15-002, *Requirements for NDI Procedure Development, Validation, and Verification for Aircraft Structural Inspections during Operations and Support Phase*, 24 April 2015

(Added-AFMC) NAS 410, *NAS [National Aerospace Standard] Certification & Qualification of Nondestructive Test Personnel*, 30 October 2020

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TO 00-25-252, *Aeronautical Equipment Welding,* 27 September 2017

TO 1-1-8, *Application and Removal of Organic Coatings, Air and Space and Non-Air and Space Equipment,* 25 October 2019

TO 1-1A-1, *General Manual for Structural Repair,* 15 January 2016

TO 1-1A-9, *Aerospace Metals - General Data and Usage Factors,* 24 January 2020

TO 1-1-686, *Desert Storage Preservation and Process Manual for Aircraft, Aircraft Engines, and Aircraft Auxiliary Power Unit Engines,* 9 April 2019

TO 1-1-691, *Aircraft Weapon Systems Cleaning and Corrosion Control,* 8 November 2019

TO 1-1-694, *Application and Removal of Low Observable Coatings on Aerospace Equipment,* 6 May 2019

TO 1-1-700, *Corrosion Prevention and Control for Ground Communications Equipment,* 9 June 2018

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TO 33B-1-2, *Nondestructive Inspection General Procedures and Process Controls,* 15 August 2019

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TO 34W4-1-5, *Operator Manual Welding Theory and Application,* 18 April 2015

TO 34-1-10, *Fundamentals of Ordnance Corps Machine Tools,* 22 April 2016

TO 35-1-3, *Corrosion Prevention, Painting and Marking of USAF Support Equipment,* 2 December 2019

AFI 62-601, *USAF Airworthiness,* 11 June 2010

(Added-AFMC) AFSCI 20-114, *Qualification of Nondestructive Inspection Personnel*, 2 July 2019

Prescribed Forms

None

Adopted Forms

Air Force Form 847, *Recommendation for Change of Publication*

Abbreviations and Acronyms

(Added-AFMC) ABDR—Aircraft Battle Damage Repair

(Added-AFMC) ACO—Advanced Composites Office

(Added-AFMC) AETC—Air Education and Training Command

(Added-AFMC) AF—Air Force

(Added-AFMC) AFCEC—Air Force Civil Engineering Center

(Added-AFMC) AFCFM—Air Force Career Field Manager

(Added-AFMC) AFCPAB—Air Force Corrosion Prevention Advisory Board

(Added-AFMC) AFCPCO—Air Force Corrosion Prevention and Control Office

AFI—Air Force Instruction

(Added-AFMC) AFLCMC—Air Force Life Cycle Management Center

AFMC—Air Force Materiel Command

(Added-AFMC) AFNDIO—Air Force Nondestructive Inspection Office

AFPD—Air Force Policy Directive

(Added-AFMC) AFRL—Air Force Research Laboratory

(Added-AFMC) AFSC—Air Force Specialty Code

(Added-AFMC) AFSC—Air Force Sustainment Center

(Added-AFMC) AFSMO—Air Force Structural Management Office

(Added-AFMC) ALC—Air Logistics Complex

(Added-AFMC) AM—Additive Manufacturing

(Added-AFMC) AMCOM—Aviation and Missile Command

(Added-AFMC) AOR—Area of Responsibility

(Added-AFMC) AS—Allowance Standard

(Added-AFMC) ASCCWG—Aerospace Systems Corrosion Control Working Group

(Added-AFMC) ASIP—Aircraft Structural Integrity Program

(Added-AFMC) CAD—Computer Aided Design

(Added-AFMC) CAM—Computer Aided Manufacturing

(Added-AFMC) CCPE—Corrosion Control and Prevention Executive

(Added-AFMC) CDC—Career Development Course

(Added-AFMC) C-E-M—Communication, Electronics and Meteorological

(Added-AFMC) CNC—Computer Numeric Control

(Added-AFMC) CPAB—Corrosion Prevention Advisory Board

(Added-AFMC) CPC—Corrosion Prevention and Control

(Added-AFMC) CTIM—Corrosion Technical Interchange Meeting

(Added-AFMC) DAFI—Department of the Air Force Instruction

(Added-AFMC) DAU—Defense Acquisition University

(Added-AFMC) ESG—Executive Steering Group

(Added-AFMC) EWG—Executive Working Group

(Added-AFMC) FYDP—Future Years Defense Program

(Added-AFMC) IG—Inspector General

(Added-AFMC) IPT—Integrated Product Team

(Added-AFMC) IRT—Independent Review Team

(Added-AFMC) LO—Low Observable

(Added-AFMC) LOIP—Low Observables Integrity Program

(Added-AFMC) M&P—Materials & Processes

MAJCOM—Major Command

(Added-AFMC) MFTO—Metal Fabrication & Technologies Office

MIL-HDBK—Military Handbook

MIL-STD—Military Standard

(Added-AFMC) MT—Metals Technology

(Added-AFMC) NAVAIR—Naval Air Systems Command

(Added-AFMC) NDE—Nondestructive Evaluation

(Added-AFMC) NDI—Nondestructive Inspection

(Added-AFMC) NDIT—Nondestructive Inspection Team

(Added-AFMC) OEM—Original Equipment Manufacturer

(Added-AFMC) OPR—Office of Primary Responsibility

(Added-AFMC) OSD—Office of the Secretary of Defense

(Added-AFMC) OSS&E—Operational Support, Sustainability and Engineering

(Added-AFMC) PEO—Program Executive Officer

(Added-AFMC) PGM—Product Group Manager

(Added-AFMC) PIWG—Product Improvement Working Group

(Added-AFMC) PM—Program Manager

(Added-AFMC) QPL—Qualified Product List

(Added-AFMC) R&D—Research and Development

(Added-AFMC) RDT&E—Research Development Testing and Evaluation

(Added-AFMC) RSO—Rapid Sustainment Office

SAF—Secretariat Offices within the Headquarters Air Force

(Added-AFMC) SAV—Staff Assistance Visit

(Added-AFMC) SCMS—Supply Chain Management Squadron

(Added-AFMC) SM—Structural Maintenance

(Added-AFMC) SME—Subject Matter Expert

(Added-AFMC) SPM—Single Program Manager

(Added-AFMC) SSC—Space System Command

(Added-AFMC) STRT—Specialty Training Requirements Team

(Added-AFMC) TCTO—Time Compliant Technical Orders

TO—Technical Order

(Added-AFMC) U&TW—Utilization and Training Workshops

USC—United States Code

(Added-AFMC) USSF—United States Space Force

Terms

Airworthiness—The property of an air system configuration to safely attain, sustain, and complete flight in accordance with approved usage limits. (AFI 62-601, *USAF Airworthiness*).

Middle Tier of Acquisition pathway—The Middle Tier of Acquisition pathway is for capabilities that have a level of maturity to allow them to be rapidly prototyped within an acquisition program or fielded within 5 years of Middle Tier of Acquisition program start. The Middle Tier of Acquisition pathway may be used to accelerate capability maturation before transitioning to another acquisition pathway or may be used to minimally develop a capability before rapid fielding. See DoDI 5000.80, *Operation of the Middle Tier of Acquisition.*

Structural Disciplines—For the purpose of this instruction, the term "structural disciplines" refers to advanced composites, corrosion prevention and control, low observables supportability, metals technology, and nondestructive inspection.

Tailoring—The manner in which certain core issues (program definition, program structure, program design, program assessments, and periodic reporting) are addressed in a particular program. The Milestone Decision Authority (MDA) seeks to minimize the time it takes to satisfy an identified need consistent with common sense, sound business management practice, applicable laws and regulations, and the time sensitive nature of the requirement itself. Tailoring may be applied to various aspects of the acquisition process, including program documentation, acquisition phases, the time and scope of decision reviews, Supportability Analysis, and decisions levels consistent with all applicable statutory requirements.

Attachment 2

AIR FORCE STRUCTURAL MANAGEMENT TECHNICAL ORDERS (TO)

Table A2.1. Air Force Structural Management Technical Orders (TO).

TO 00-25-224 *Welding High Pressure and Cryogenic Systems*
TO 00-25-252 *Aeronautical Equipment Welding*
TO 1-1-8 *Application and Removal of Organic Coatings, Air and Space and Non-Air and Space Equipment*
TO 1-1A-1 *General Manual for Structural Repair*
TO 1-1A-9 *Aerospace Metals General Data and Usage Factors*
TO 1-1-686 *Desert Storage Preservation and Process Manual for Aircraft, Aircraft Engines, and Aircraft Auxiliary Power Unit Engines*
TO 1-1-690 *General Advanced Composite Repair Processes Manual*
TO 1-1-691 *Aircraft Weapon Systems Cleaning and Corrosion Control*
TO 1-1-694 *Application and Removal of Low Observable Coatings on Aerospace Equipment*
TO 1-1-700 *Corrosion Prevention and Control for Ground Communications Equipment*
TO 32-1-101 *Use and Care of Hand Tools and Measuring Tools*
TO 33B-1-1 *Non-Destructive Inspection Methods, Basic Theory*
TO 33B-1-2 *Nondestructive Inspection General Procedures and Process Controls*
TO 34-1-3 *Technical Manual Inspection and Maintenance of Machinery and Shop Equipment*
TO 34W4-1-5 *Operator Manual Welding Theory and Application*
TO 34-1-10 *Fundamentals of Ordnance Corps Machine Tools*
TO 35-1-3 *Corrosion Prevention, Painting and Marking of USAF Support Equipment*

Note: Referenced In Paragraph 2.4.3.5.

Attachment 3 (Added-AFMC)

AFMC/AFSMO MANAGED TOS

Table A3.1. (AFMC) AFMC/AFSMO Managed TOs.

TO Number	TO Title	CPCO	NDIO	MFTO	ACO	LO
00-25-224	Welding High Pressure and Cryogenic Systems			X		
00-25-252	Aeronautical Equipment Welding			X		
00-80C-1	Crash Damaged, Disabled Aircraft Recovery Manual				X	
1-1-8	Application and Removal of Organic Coatings, Aerospace and Non-Aerospace Equipment	X				
1-1A-9	Aerospace Metals General Data and Usage Factors			X		
1-1-686	Desert Storage Preservation and Process Manual for Aircraft, Aircraft Engines, and Aircraft Auxiliary Power Unit Engines	X				
1-1-689-1	Cleaning and Corrosion Control: Volume I, Corrosion Program and Corrosion Theory	X				
1-1-689-3	Cleaning and Corrosion Control: Volume III, Avionics and Electronics	X				
1-1-689-5	Cleaning and Corrosion Control: Volume V, Consumable Materials and Equipment for Avionics	X				
1-1-690	General Advanced Composite Repair Processes Manual				X	
1-1-691	Cleaning and Corrosion Prevention and Control, Aerospace and Non-Aerospace Equipment	X				
1-1-694	Application and Removal of Low Observable Coatings on Aerospace Equipment					X
1-1-700	Corrosion Prevention and Control for Ground Communications Equipment, Electronics and Meteorological (C-E-M) Equipment and its Protective Shelters, Vans, and Radomes	X				
32-1-101	Use and Care of Hand Tools and Measuring Tools			X		
33B-1-1	NDI Methods, Basic Theory		X			

33B-1-2	Nondestructive Inspection General Procedures and Process Controls		X			
34-1-3	Inspection and Maintenance of Machinery and Shop Equipment			X		
34W4-1-5	Welding Theory and Application			X		
34-1-10	Fundamentals of Ordnance Corps Machine Tools			X		
35-1-3	Corrosion Prevention and Control, Cleaning, Painting and Marking of USAF Support Equipment (SE)	X				
34A-1-1	Additive Manufacturing Qualification of Technicians, Machines, and Facilities Additive			X		
34A-2-1	Metals Additive Manufacturing General Procedures and Process Controls			X		
34A-3-1	Polymers Additive Manufacturing General Procedures and Process Controls			X		